

Communication Systems

Question1

The heights of transmitting and receiving antennas are respectively $\frac{1}{20000}$ and $\frac{1}{80000}$ times the radius of the

Earth. The maximum distance between these two antennas for satisfactory communication in line of sight mode is

(Radius of the Earth = 6.4×10^6 m)

AP EAPCET 2025 - 26th May Morning Shift

Options:

A.

48 km

B.

96 km

C.

320 km

D.

192 km

Answer: B

Solution:

$$H_t = \frac{1}{20000} R = \frac{R}{20000}$$

$$H_r = \frac{R}{80000}$$

Maximum distance

$$\begin{aligned} d &= \sqrt{2RH_t} + \sqrt{2RH_r} \\ &= \sqrt{2R \times \frac{R}{20000}} + \sqrt{2R \times \frac{R}{80000}} \\ &= \frac{R}{100} + \frac{R}{200} = \frac{6.4 \times 10^6}{100} + \frac{6.4 \times 10^6}{200} \\ &= 6.4 \times 10^4 + 3.2 \times 10^4 \\ &= 9.6 \times 10^4 \text{ m} = 96 \times 10^3 \text{ m} = 96 \text{ km} \end{aligned}$$

Question2

If the ratio of the maximum and minimum amplitudes of an amplitude modulated wave is 7 : 3, then the modulation index is

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Options:

A.

0.6

B.

0.7

C.

0.4

D.

0.3

Answer: C

Solution:

Modulation index is given by

$$\begin{aligned}\mu &= \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \\ &= \frac{\frac{A_{\max}}{A_{\min}} - 1}{\frac{A_{\max}}{A_{\min}} + 1} \\ &= \frac{\frac{7}{3} - 1}{\frac{7}{3} + 1} = \frac{\frac{4}{3}}{\frac{10}{3}} = 0.4\end{aligned}$$

Question3

Ionosphere acts as a reflector for the frequency range of

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Options:

A.

3 – 30kHz

B.

3 – 30MHz

C.

3 – 30 Hz

D.

3 – 30GHz

Answer: B

Solution:

The ionosphere reflects **high frequency (HF)** radio waves, which correspond to the range:

3 MHz to 30 MHz

This property allows long-distance radio communication (sky wave propagation).

 **Correct answer:**

Option B: 3 – 30 MHz

Question4

If the maximum and minimum amplitudes of a modulated wave are 25 V and 5 V respectively, then the modulation index is

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Options:

A.

$$\frac{1}{5}$$

B.

$$\frac{1}{3}$$

C.

$$\frac{3}{2}$$

D.

$$\frac{2}{3}$$

Answer: D

Solution:

$$A_{\max} = 25 \text{ V}$$

$$A_{\min} = 5 \text{ V}$$

$\therefore$ Modulation index

$$\begin{aligned}\mu &= \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} \\ &= \frac{25 - 5}{25 + 5} = \frac{20}{30} = \frac{2}{3}\end{aligned}$$

Question5

For transmitting a signal of frequency 1000 kHz , the minimum length of the antenna is

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Options:

A.

30 m

B.

50 m

C.

75 m

D.

1500 m

Answer: C

Solution:

Step 1: Write down the given data

$$f = 1000 \text{ kHz} = 1 \times 10^6 \text{ Hz}$$

Speed of electromagnetic wave in free space,

$$c = 3 \times 10^8 \text{ m/s}$$

Step 2: Find the wavelength

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{1 \times 10^6} = 300 \text{ m}$$

Step 3: Relation between antenna length and wavelength

For efficient radiation, the **minimum antenna length** (usually a quarter-wave antenna) is:

$$L = \frac{\lambda}{4}$$

Step 4: Calculate L

$$L = \frac{300}{4} = 75 \text{ m}$$

 **Answer: Option C — 75 m**

Question6

If in an amplitude modulated wave, the maximum amplitude is 14 V and the modulation index is 0.4 , then the amplitude of the carrier wave is

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Options:

A.

4 V

B.

8 V

C.

12 V

D.

10 V

Answer: D

Solution:

$$\begin{aligned}A_{\max} &= A_m + A_c \\&= \mu A_c + A_c = (\mu + 1)A_c \\ \Rightarrow A_c &= \frac{A_{\max}}{\mu + 1} = \frac{14}{0.4 + 1} = \frac{14}{1.4} = 10 \text{ V}\end{aligned}$$

Question7

In amplitude modulation of waves, the maximum amplitude is 30 mV and minimum amplitude is 5 mV , then the modulation index is

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Options:

A.

$$\frac{4}{7}$$

B.

$$\frac{3}{7}$$

C.

$$\frac{5}{7}$$

D.

$$\frac{2}{7}$$

Answer: C

Solution:

Given:

- Maximum amplitude $A_{\max} = 30 \text{ mV}$
- Minimum amplitude $A_{\min} = 5 \text{ mV}$

Formula for modulation index m :

$$m = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

Substitute the values:

$$m = \frac{30-5}{30+5} = \frac{25}{35} = \frac{5}{7}$$

 **Answer: Option C — $\frac{5}{7}$**

Question8

If the sum of heights of transmitting and receiving antennas in line of sight of communication is ' h ' then the height of receiving

antenna, to have the range maximum is

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Options:

A.

$$\frac{h}{2}$$

B.

$$\frac{h}{4}$$

C.

$$2h$$

D.

$$\frac{2h}{3}$$

Answer: A

Solution:

The range of line of sight communication between two antennas is given by

$$x = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Here, $h_T + h_R = h$

Let $h_T = H$, then

$$h_R = h - H$$

$$\therefore x = \sqrt{2R}[\sqrt{H} + \sqrt{h - H}]$$

$$\frac{dx}{dH} = \sqrt{2R} \left[\frac{1}{2\sqrt{H}} + \frac{1}{2\sqrt{h - H}}(-1) \right] = 0$$

$$\Rightarrow \frac{1}{2\sqrt{H}} = \frac{1}{2\sqrt{h - H}}$$
$$H = h - H$$

$$\Rightarrow H = \frac{h}{2}$$

Question9

Which one of the following is not classified as pulse modulation?

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Options:

- A. Pulse duration modulation
- B. Pulse amplitude modulation
- C. Pulse band modulation
- D. Pulse position modulation

Answer: C

Solution:

The correct answer is Option C: Pulse band modulation.

Here's a brief explanation of why:

Pulse duration modulation (often called Pulse Width Modulation, PWM) varies the width or duration of the pulse.

Pulse amplitude modulation (PAM) varies the amplitude of the pulse.

Pulse position modulation (PPM) varies the position of the pulse within a time frame.

Pulse band modulation is not a standard or recognized method in the classification of pulse modulation techniques.

Therefore, Option C is the one that does not belong.

Question10

The maximum distance between the transmitting and receiving antennas for satisfactory communication in line of sight mode is 57.6 km . If the height of the receiving antenna is 80 m , the height of the transmitting antenna is (radius of earth = 6.4×10^6 m)

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Options:

A. 28.8 m

B. 51.2 m

C. 25.6 m

D. 14.4 m

Answer: B

Solution:

Height of receiving antenna = 80 m = h_t

Maximum distance between

transmitting and receiving antenna in

line of sight mode = $d_m = 57.6 \text{ km} = 57600 \text{ m}$

Radius of earth = $6.4 \times 10^6 \text{ m}$

Let height of transmitting antenna $a = h_t$

We know that, maximum line of sight distance between two antennas

$$\begin{aligned}d_m &= \sqrt{2Rh_T} + \sqrt{2Rh_t} \\ \Rightarrow 57600 &= \sqrt{2 \times 6.4 \times 10^6 \times h_T} + \sqrt{2 \times 6.4 \times 10^6 \times 80} \\ &= \sqrt{12.8 \times 10^6 \times h_T} + \sqrt{1024 \times 10^6} \\ &= \sqrt{12.8 \times 10^6 \times h_T} + 32000 \\ \Rightarrow \sqrt{12.8 \times 10^6 \times h_T} &= 25600 = 256 \times 10^2 \\ \Rightarrow 128 \times 10^6 \times h_T &= 65536 \times 10^4 \\ \Rightarrow h_T &= \frac{65536 \times 10^4}{12.8 \times 10^6} \\ &= \frac{5120}{10^2} = 51.20 \\ h_T &= 51.20 \text{ m}\end{aligned}$$

Question 11

An information signal of frequency 10 kHz is modulated with a carried wave of frequency 3.62 MHz. The upper side and lower side frequencies are

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Options:

- A. 3650 kHz and 3590 kHz
- B. 3620 kHz and 3600 kHz
- C. 3610 kHz and 3580 kHz
- D. 3600 kHz and 3620 kHz

Answer: B

Solution:

Given:

Signal frequency, $v_s = 10$ kHz

Carrier wave frequency, $v_c = 3.61$ MHz = 3610 kHz

We can calculate the side band frequencies using these formulas:

Upper Side Band Frequency:

The formula for the upper side band frequency is:

$$v_u = v_c + v_s$$

Substitute the given values:

$$v_u = 3610 + 10$$

$$v_u = 3620 \text{ kHz}$$

Lower Side Band Frequency:

The formula for the lower side band frequency is:

$$v_l = v_c - v_s$$

Substitute the given values:

$$v_l = 3610 - 10$$

$$v_l = 3600 \text{ kHz}$$

Therefore, the upper side band frequency is 3620 kHz, and the lower side band frequency is 3600 kHz.

Question12

In amplitude modulation, the amplitude of the carrier wave is 10 V and the amplitude of one of the side bands is 2 V . Then, the modulation index is

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Options:

A. 0.4

B. 0,6

C. 0.7

D. 0.5

Answer: A

Solution:

To determine the modulation index in amplitude modulation:

Given:

Carrier amplitude, $A_c = 10 \text{ V}$

Side band amplitude, $A_s = 2 \text{ V}$

The relationship between A_c and A_s is defined by the formula:

$$A_s = \frac{A_c \times m}{2}$$

Here, m represents the modulation index. Rearranging the formula to solve for m , we have:

$$m = \frac{2A_s}{A_c}$$

Substituting the given values:

$$m = \frac{2 \times 2}{10} = 0.4$$

Thus, the modulation index is 0.4.

Question13

A carrier is simultaneously modulated by two sine waves with modulation indices of 0.3 and 0.4 , then the total modulation index is

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Options:

- A. 1
- B. 0.12
- C. 0.5
- D. 0.7

Answer: C

Solution:

The effective modulation index, when a carrier is modulated by two sine waves, can be calculated using the formula:

$$\mu = \sqrt{\mu_1^2 + \mu_2^2}$$

Where:

$$\mu_1 = 0.3$$

$$\mu_2 = 0.4$$

To find the total modulation index:

$$\mu = \sqrt{(0.3)^2 + (0.4)^2}$$

Calculating inside the square root:

$$= \sqrt{0.09 + 0.16}$$

$$= \sqrt{0.25}$$

$$= 0.5$$

Therefore, the total modulation index is 0.5.

Question14

If the maximum and minimum voltages of an AM wave are $V_{\max}$ and $V_{\min}$ respectively. Then, the modulation factor m is

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Options:

A. $(V_{\max} + V_{\min}) / (V_{\max} \cdot V_{\min})$

$$\text{B. } (V_{\max} - V_{\min}) / (V_{\max} + V_{\min})$$

$$\text{C. } 2V_{\max} - V_{\min} / (V_{\max} + V_{\max})$$

$$\text{D. } (V_{\max} + V_{\min}) / (V_{\max} - V_{\min})$$

Answer: B

Solution:

Given:

Maximum voltage of an AM wave: $V_{\max}$

Minimum voltage of an AM wave: $V_{\min}$

Amplitude of the modulating signal is calculated as:

$$A_m = \frac{V_{\max} - V_{\min}}{2}$$

Amplitude of the carrier signal is calculated as:

$$A_C = \frac{V_{\max} + V_{\min}}{2}$$

The **modulation index** or **modulation factor** m is defined as the ratio of the amplitude of the modulating signal to the amplitude of the carrier signal:

$$m = \frac{A_m}{A_C}$$

Substituting the expressions for A_m and A_C , the modulation index becomes:

$$m = \frac{\frac{V_{\max} - V_{\min}}{2}}{\frac{V_{\max} + V_{\min}}{2}} \Rightarrow m = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}$$
